CSE 3231 – Computer Networks

Spring 2026

Final

Date: **April 27, 2026**

Due Date: May 4th, by midnight (firm deadline)

Student Name: ______________________________

ID: ______________________________

Team Name: ______________________________

(if applicable)

No Restrictions

The use of AI is permitted as a research tool, but NOT permitted for producing any part of the deliverables.

Professor:

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Task Description

The goal of this homework is to practice the basic principles of TCP congestion control and flow control. You are asked to produce a short document explaining how TCP works, with special attention to TCP's congestion and flow control mechanisms. You will also need to create a simple simulation program that demonstrates how TCP's congestion and flow control algorithms work. Part of the assignment is your determination of the best way to illustrate those functionalities in your program, so no further specification will be provided for the simulation.

Students can work independently or in teams of up to 3 people on all deliverables. Each student must individually upload the submission on Canvas by the deadline, and the same submission files can be uploaded by all team members.

If you work as a team, please give your team a name and ensure you include the team's name in all deliverables (including report and code), in addition to your own name and student ID.

Deliverables

1. **Technical report** (max. 4 pages) as a PDF (or MS Word) file, explaining how TCP works, with special attention to the congestion and flow control mechanisms.
2. **Your simulation code** (any language)
3. **Code report** (max 2 pages), in the form of a separate PDF or MSWord file, including a description of how your application works, how to run it, and a printout of the output of your program.

Important Requirements:

- Ensure that your name, student ID, and team name (if applicable) are included on the cover of each deliverable.
- Compress all three deliverables as a **single file** (student-name.zip or team-name.zip), and upload that single file as your deliverable.